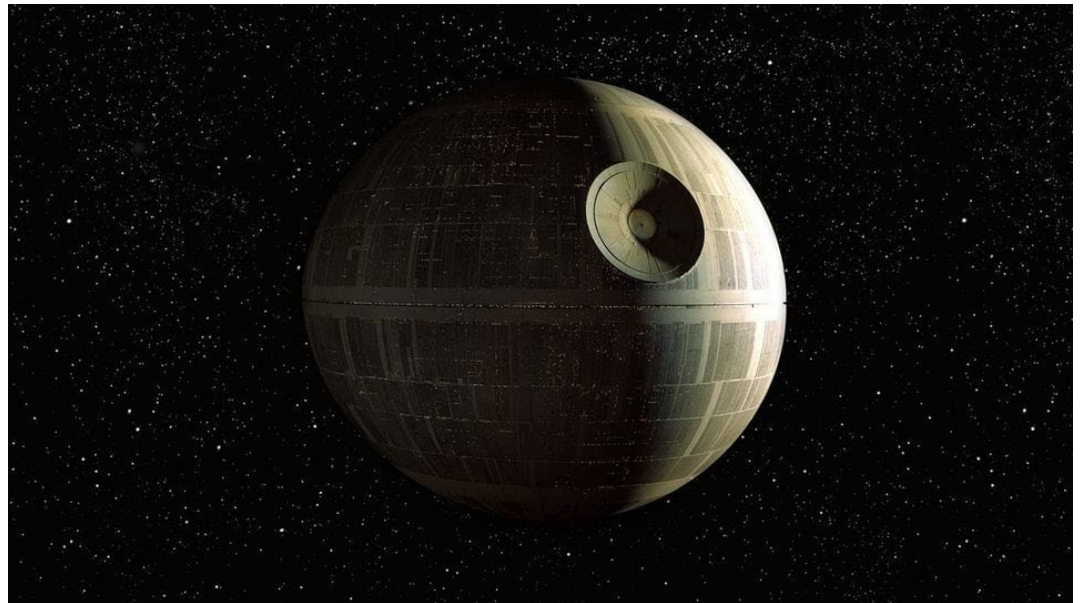


Cylinders and Quadric Surfaces

Outline:

1. Recognizing and graphing cylinders and quadric surfaces
2. 3D curves of intersection



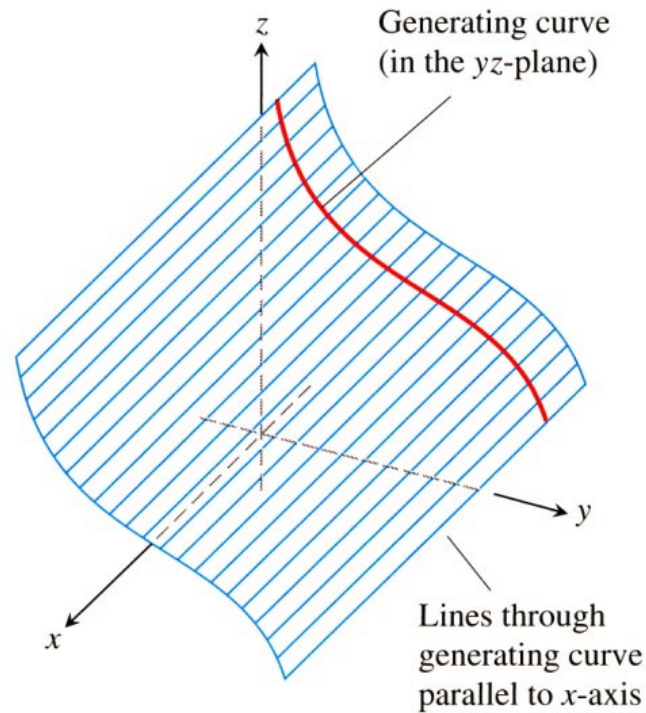
The Death Star was the Empire's ultimate weapon: a moon-sized space station with the ability to destroy an entire planet ...

<https://www.starwars.com/databank/death-star>



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Cylinders

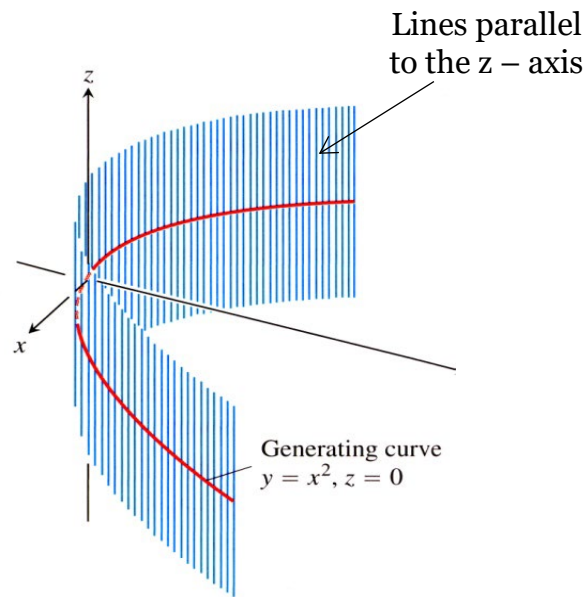


A cylinder is a surface generated by moving a straight line along a given planar curve while holding the line parallel to a given fixed line

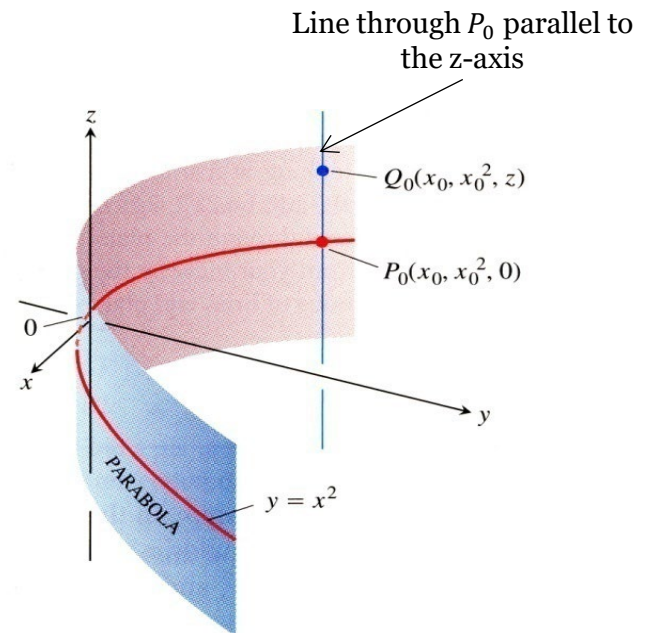


Cylinders

Equation: $y = x^2$



→ Parabolic Cylinder



Every point in the cylinder has coordinates:

$$(x_0, x_0^2, z)$$



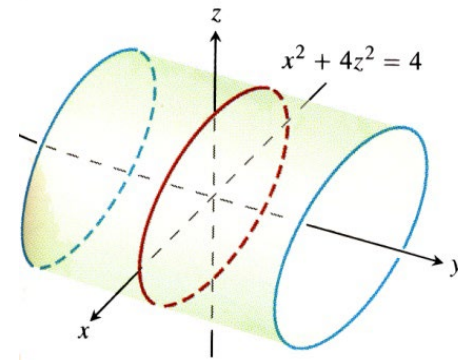
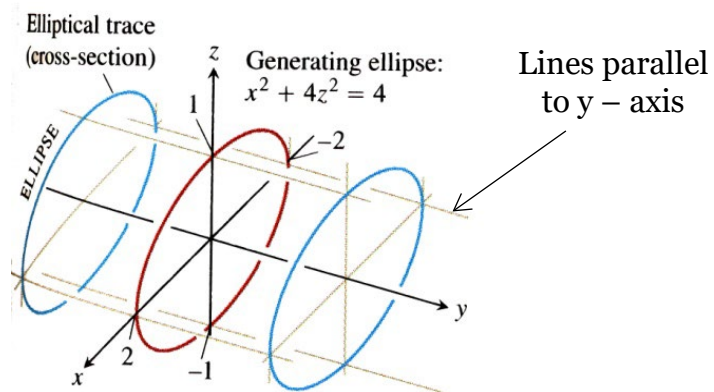
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Cylinders

General Form: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

→ Elliptical Cylinder

Example: $x^2 + 4z^2 = 4$



Every point in the cylinder has coordinates:

$$(x_0, y, z_0)$$

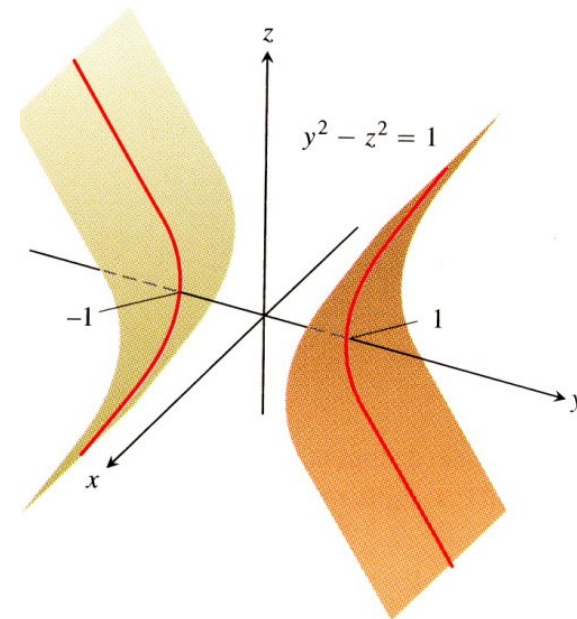
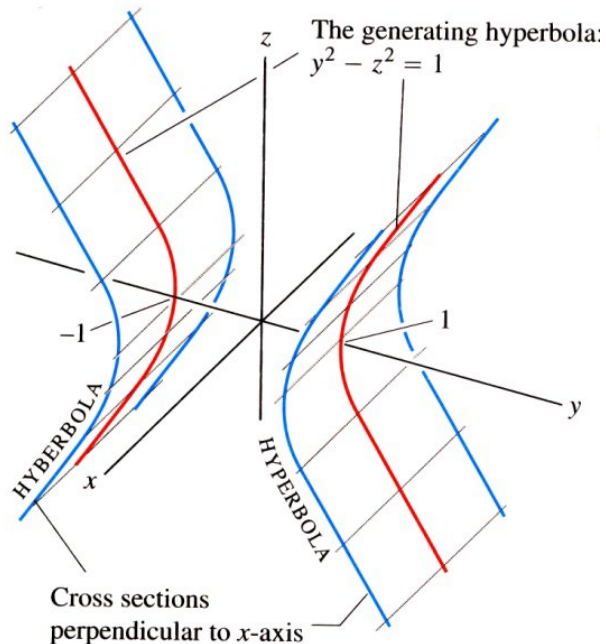


Cylinders

General Form: $\frac{y^2}{a^2} - \frac{z^2}{b^2} = 1$

Example: $y^2 - z^2 = 1$

→ Hyperbolic Cylinder



Every point in the cylinder has coordinates:

$$(x, y_0, z_0)$$



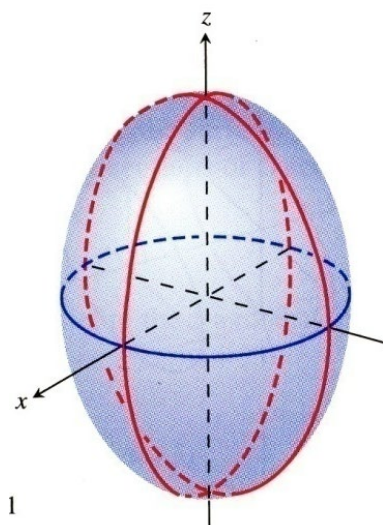
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Quadric Surfaces

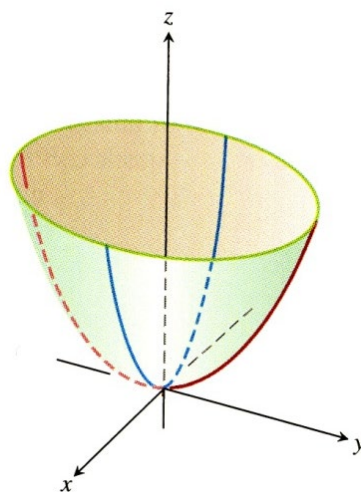
Graph of any equation that can be put into the general form:

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$$

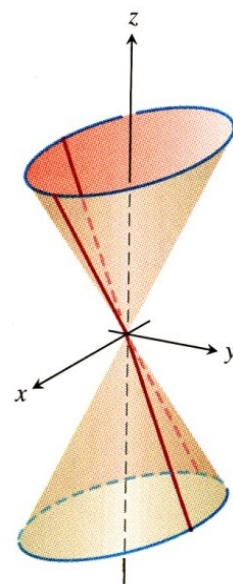
Where, A, B, ..., are constants



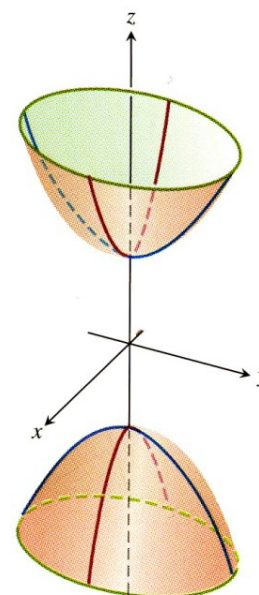
Ellipsoid



Paraboloid



Elliptical Cones



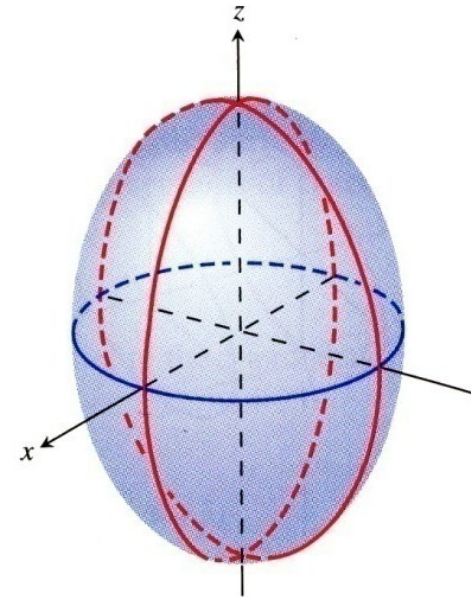
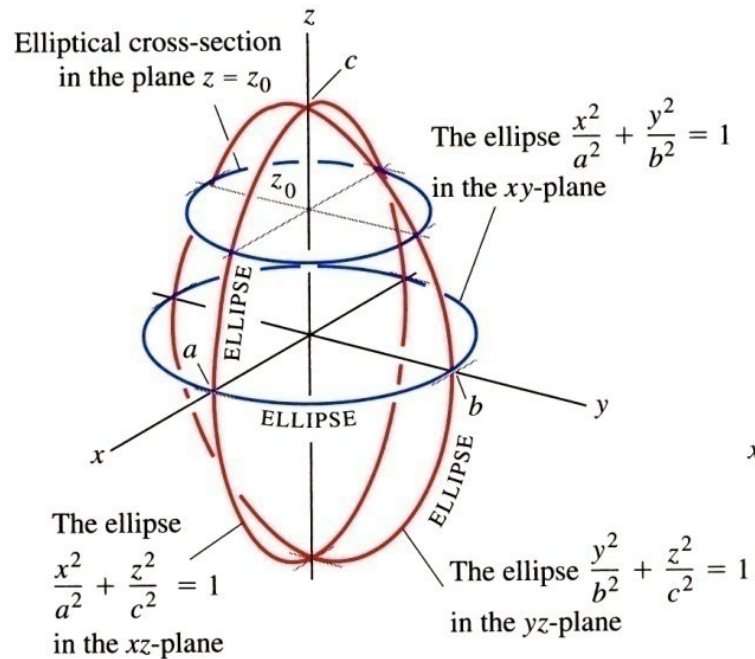
Hyperboloids



Quadric Surfaces

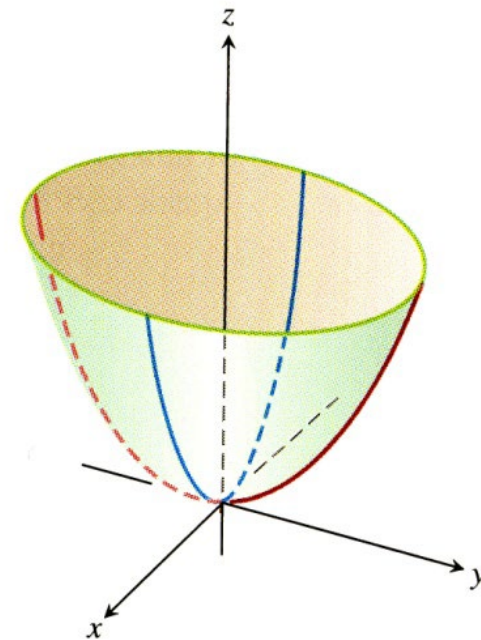
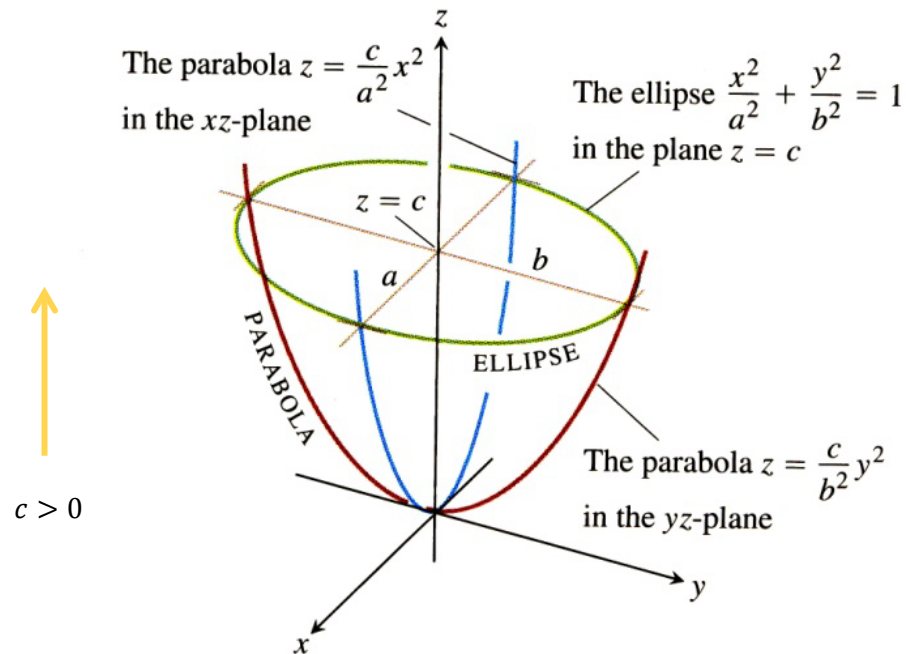
General Form: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

→ Ellipsoid



Quadric Surfaces

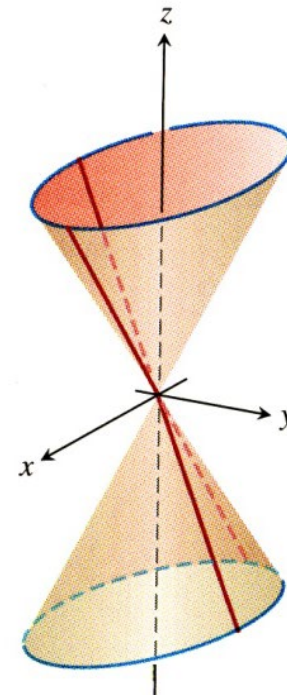
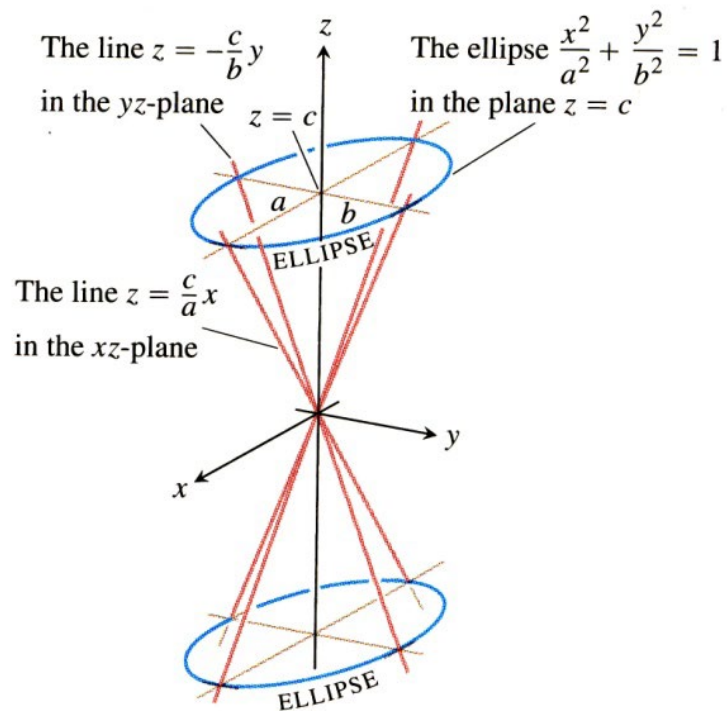
General Form: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$ $\rightarrow$ Elliptical Paraboloid



Quadric Surfaces

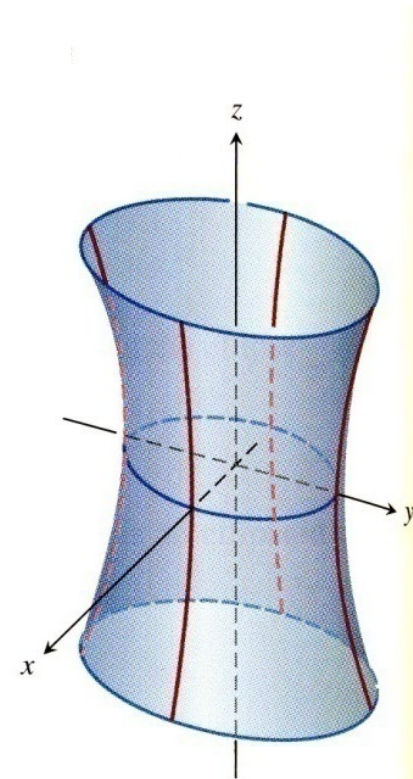
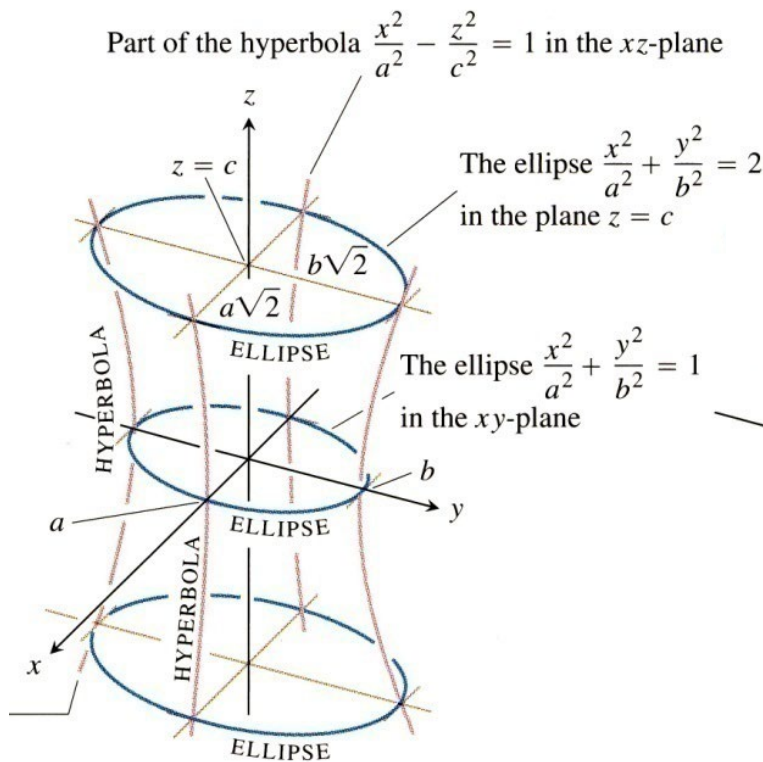
General Form: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$

→ Elliptical Cone



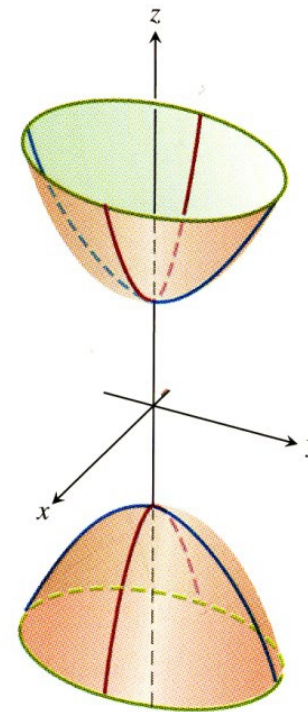
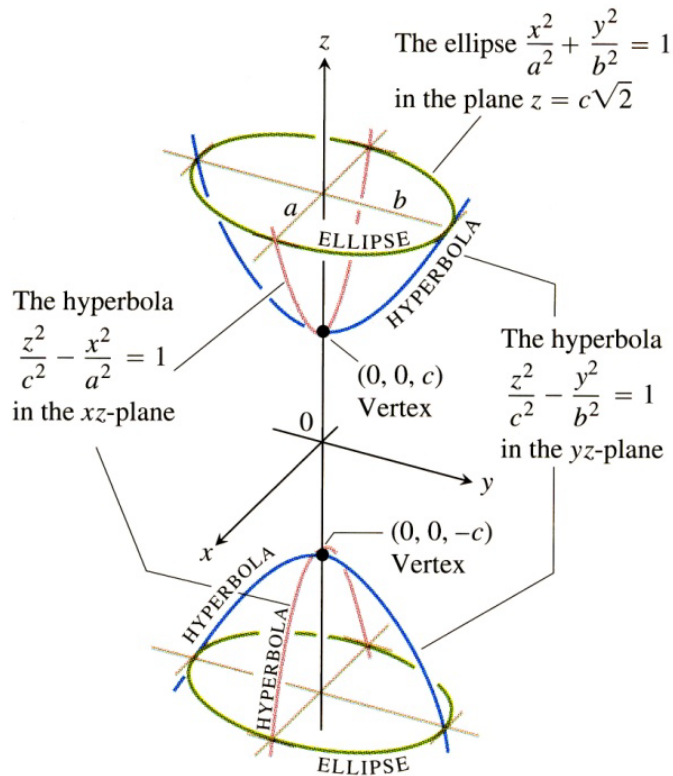
Quadric Surfaces

General Form: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ $\rightarrow$ Hyperboloid of one sheet



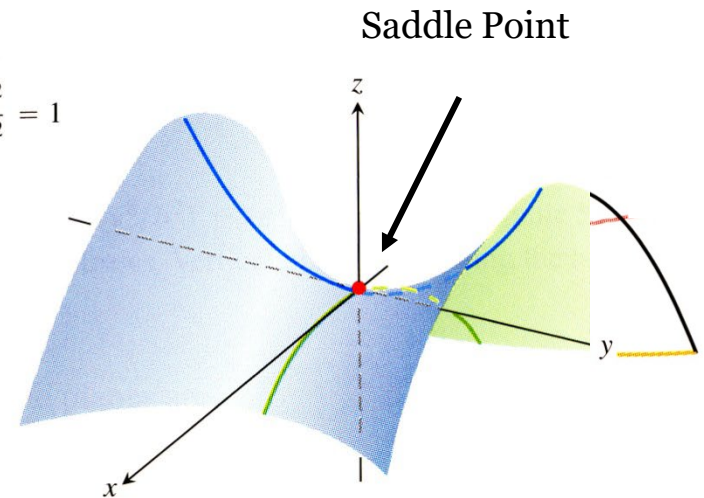
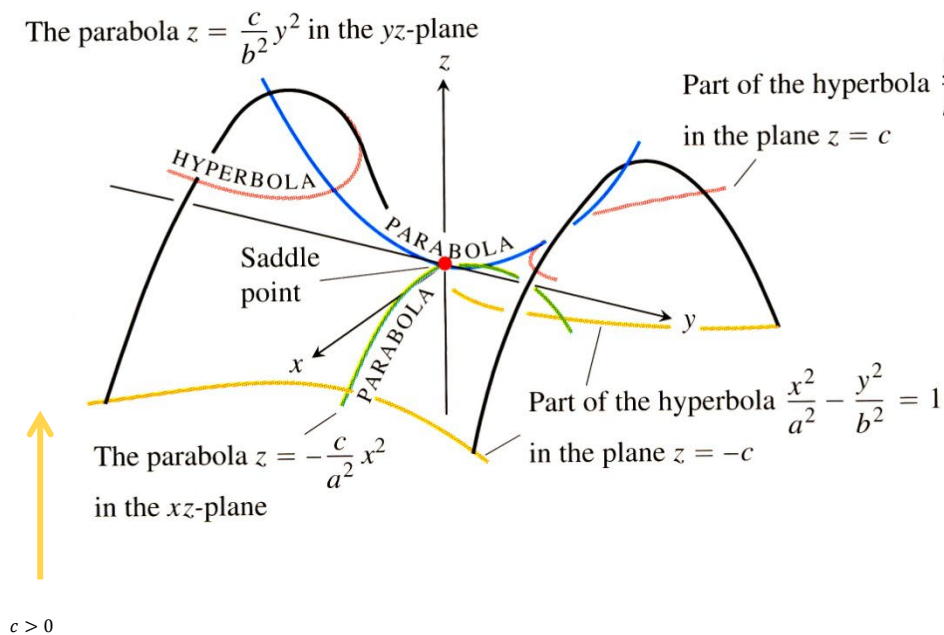
Quadric Surfaces

General Form: $\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \rightarrow \quad \text{Hyperboloid of two sheets}$



Quadric Surfaces

General Form: $\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}$ $\rightarrow$ Hyperbolic Paraboloid



3D Curves of Intersection

Problem

A particle moves along the curve represented by the intersection of the surfaces

$$z = x^2 + y^2$$

and,

$$x + z = 1.$$

Draw the curve of intersection and find the equation for the projection of the curve in the xy plane

Solution:

To draw the curve of intersection between surfaces, we need to be able to draw the surfaces and visually determine their intersection.

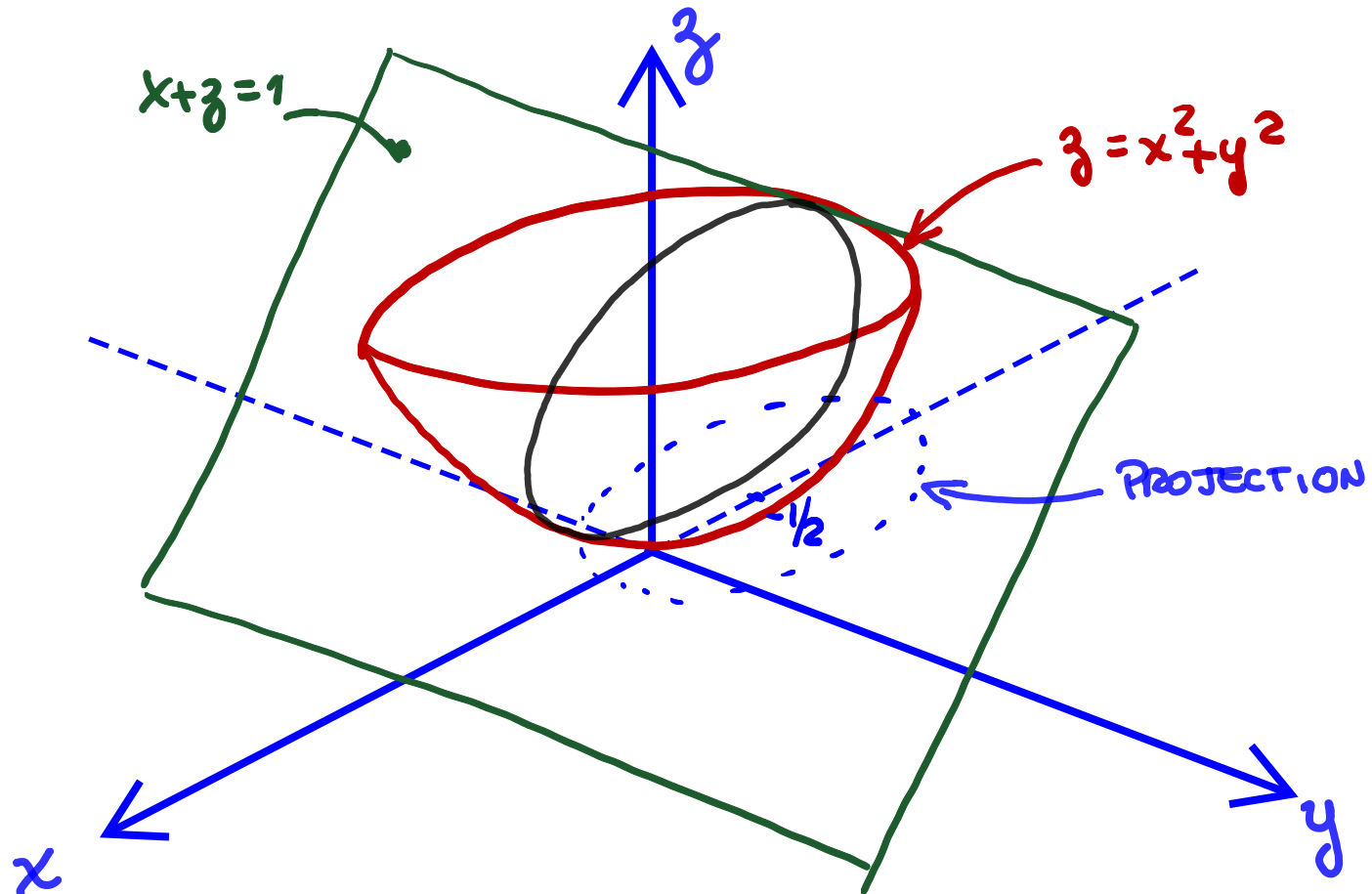
The surface $z = x^2 + y^2$ is a ... **PARABOLOID**

The surface $x + z = 1$ is a ... **PLANE**



Curve of Intersection Between Surfaces

Sketching the curve of intersection:



3D Curves of Intersection

Now that we have sketched the curve, let us determine its **projection on the $x - y$ plane**:

The projection of the curve in the x - y plane can be found by equating the equations of both surfaces:

$$z = x^2 + y^2 \quad (1)$$

$$x + z = 1 \quad (2)$$

FROM (2) $\rightarrow z = 1 - x \rightarrow$ WE WANT TO ELIMINATE "z"

INTO (1) $\rightarrow (1 - x) = x^2 + y^2 \dots (3)$

RE-ARRANGING AND FACTORIZING (3),

$$x^2 + x - 1 + y^2 = 0$$



3D Curves of Intersection

$$x^2 + x + \frac{1}{4} - \frac{1}{4} + y^2 = 1$$

$$\left(x + \frac{1}{2}\right)^2 + y^2 = \frac{5}{4}$$



CIRCLE CENTERED AT $\left(-\frac{1}{2}, 0\right)$ WITH
RADIUS $\frac{\sqrt{5}}{2}$

This all looks interesting, but how do we write an expression for this curve in 3D space?



Summary

- We introduced several 3D cylinder and quadric surfaces
- We learned how to recognize them and graph them
- We talked about how curves are formed by the intersection surfaces

We will determine the representation of 3D curves using vector functions during our next lecture.



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Thank you